

Algebra I

Course Outline • Mathlify Summer Program

Course Overview

This course builds strong algebra foundations through structured instruction and consistent practice. Students develop fluency with equations, functions, and graphing while strengthening problem-solving skills needed for Geometry and Algebra II.

Course Units

Unit 1 — Foundations of Algebra

- Order of operations
- Evaluating expressions
- Properties of real numbers

Unit 2 — Solving Linear Equations & Inequalities

- One-step and multi-step equations
- Solving and graphing inequalities
- Word problems with equations

Unit 3 — Linear Functions

- Graphing in slope-intercept form
- Writing equations of lines
- Interpreting slope and rate of change

Unit 4 — Systems of Equations

- Solving systems by graphing
- Substitution and elimination methods
- Applications of systems

Unit 5 — Exponents & Polynomials

- Laws of exponents
- Adding, subtracting, and multiplying polynomials
- Factoring basic trinomials

Unit 6 — Quadratic Functions

- Graphing quadratics
- Factoring and solving quadratic equations
- Intro to completing the square and quadratic formula

Unit 7 — Radicals & Intro to Functions

- Simplifying square roots
- Intro to function notation
- Comparing linear vs quadratic models